Unit1: Ordinary Differential Equation of Higher Order

Exercise1. Classify the following equations by its kind, order, degree and linearity:

(1) $\frac{d^3y}{dx^3}+\frac{d^2y}{dx^2}+y=x$, (2) $\frac{dy}{dx}+y^2=x^2$, (3) $x^2\left(\frac{d^2y}{dx^2}\right)^3+y\left(\frac{dy}{dx}\right)^4+y^4=0$, (4) $\left(\frac{dr}{ds}\right)^3=\sqrt{\frac{d^2r}{ds^2}+1}$

(5) $\frac{\partial^2 y}{\partial t^2}=a^2\frac{\partial^2 y}{\partial x^2}$ (6) $\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}=\frac{d^2y}{dx^2}$

Ans: (1) ODE, 3, 1, Linear; (2) ODE, 1, 1, Nonlinear; (3) ODE, 2, 3, Nonlinear; (4) ODE, 2, 1, Nonlinear; (5) PDE, 2, 1, Linear; (6) ODE, 2, 2, Nonlinear

Exercise 2. Solve the following differential equations:

(1) $\frac{d^2y}{dx^2}-\frac{dy}{dx}-12y=0$ (2) $\frac{d^2y}{dx^2}+a^2y=0$ (3) $\frac{d^3y}{dx^3}-\frac{d^2y}{dx^2}+\frac{dy}{dx}-y=0$ (4) $(D-3)^3y=0$

(5) $\frac{d^2y}{dx^2}-3\frac{dy}{dx}+2y=0,\ y(0)=-1,\ y'(0)=0$ (6) $\frac{d^4y}{dx^4}-m^4y=0$ (7) $(D^4+4D^3+8D^2+8D+4)y=0$

(8) $(D^2+1)^2(D-1)^2y=0$ (9) $(D^4-4D^3+8D^2-8D+4)y=0$ (10) Find the value of y which satisfies the equation $\frac{d^2y}{dx^2}+4\frac{dy}{dx}-12y=0$, given that $y=0$ and $\frac{dy}{dx}=1$ when $x=0$.

Ans: (1) $y=c_1e^{-3x}+c_2e^{4x}$ (2) $y=c_1\cos ax+c_2\sin ax$ (3) $y=c_1e^x+(c_2\cos x+c_3\sin x)$
(4) $y=(c_1+c_2x++c_3x^2)e^{3x}$ (5) $y=e^{2x}-2e^x$ (6) $y=c_1e^{mx}+c_2e^{-mx}+c_3\cos mx+c_4\sin mx$
(7) $y=e^{-x}[(c_1+c_2x)\cos x+(c_3+c_4x)\sin x]$ (8) $y=(c_1+c_2x)\cos x+(c_3+c_4x)\sin x+(c_5+c_6x)e^x$
(9) $y=e^x[(c_1+c_2x)\cos x+(c_3+c_4x)]\sin x$ (10) $y=-\frac{1}{8}e^{-6x}+\frac{1}{8}e^{2x}$

Exercise 3. Solve the following differential equations

(1) $(D^3-12D+16)y=(e^x+e^{-2x})^2$, (2) $(D^3+1)y=3+5e^x$, (3) $y''+4y'+13y=18e^{-2x}$, $y(0)=0,\ y'(0)=4$,

(4) $(D^2+5D+6)y=e^{2x}$, given that $y=0,\ \frac{dy}{dx}=0$, when $x=0$, (5) $(D^2-p^2)y=\sinh px$,

(6) $(D^2+3D+2)y=e^{2x}+e^{-x}+\sin 2x$, (7) $\frac{d^3y}{dx^3}-3\frac{d^2y}{dx^2}+4\frac{dy}{dx}-2y=e^x+\cos x$, (8) $(D^3+a^2D)y=\sin ax$,

(9) $(D-1)^2(D^2+1)^2y=e^x+\sin^2\left(\frac{x}{2}\right)$, (10) Solve $\frac{d^2y}{dx^2}+2\frac{dy}{dx}+10y+37\sin 3x=0$ and find the value of y when $x=\frac{\pi}{2}$, if it is given that $y=3$ and $\frac{dy}{dx}=-3$, when $x=0$,

(11) $(D^2-4D+1)y=\cos x\cos 2x+\sin^2 x$, (12) $y''+4y'+5y+2\cosh x=0$, $y(0)=0,\ y'(0)=1$,

(13) $(D-2)^2y=8(e^{2x}+\sin 2x+x^2)$, (14) $(D^2-4D-5)y=e^{2x}+3\cos(4x+3)$

Ans: (1) $y=(c_1+c_2x)e^{2x}+c_3e^{-4x}+\frac{x^2e^{2x}}{12}+\frac{2e^{-x}}{27}+\frac{xe^{-4x}}{36}$,

(2) $y=c_1e^{-x}+e^{\frac{x}{2}}\left(c_2\cos\frac{\sqrt{3}}{2}x+c_3\sin\frac{\sqrt{3}}{2}x\right)+3+\frac{5}{2}e^x$

(3) $y=e^{-2x}\left(\frac{4}{3}\sin 3x-2\cos 3x\right)+2e^{-2x}$, (4) $y=\frac{1}{5}e^{-3x}-\frac{1}{4}e^{-2x}+\frac{1}{20}e^{2x}$

(5) $y=(c_1e^{px}+c_2e^{-px})+\frac{x}{2p}\cosh px$ (6) $y=(c_1e^{-x}+c_2e^{-2x})+\frac{1}{12}e^{2x}+xe^x-\frac{1}{20}(3\cos 2x+\sin 2x)$

(7) $y = c_1 e^x + e^x(c_2 \cos x + c_3 \sin x) + xe^x + (1/10)(3\sin x + \cos x)$

(8) $y = c_1 + c_2 \cos ax + c_3 \sin ax - \dfrac{x}{2a^2}\sin ax$

(9) $y = (c_1 + c_2 x)e^x + (c_3 + c_4 x)\cos x + (c_5 + c_6 x)\sin x + \dfrac{1}{2} - \dfrac{1}{32}x^2 \sin x + \dfrac{1}{8}x^2 e^x$

(10) $y = e^{-x}(c_1 \cos 3x + c_2 \sin 3x) + 6\cos 3x - \sin 3x$, $y = 1 + e^{-\pi/2}$

(11) $y = (c_1 e^{(2+\sqrt{3})x} + c_2 e^{(2-\sqrt{3})x}) + \dfrac{1}{2}\left(1 - \dfrac{1}{4}\sin x - \dfrac{1}{73}(3\cos 2x + 8\sin 2x) - \dfrac{1}{52}(3\sin 3x + 2\cos 3x)\right)$

(12) $y = \dfrac{3}{5}e^{-2x}(\cos x + 3\sin x) - \dfrac{e^x}{10} - \dfrac{e^{-x}}{2}$, (13) $y = (c_1 + c_2 x)e^{2x} + 4x^2 e^{2x} + \cos 2x + 2x^2 + 4x + 3$

(14) $y = c_1 e^{-x} + c_2 e^{5x} - \dfrac{1}{9}e^{2x} + \dfrac{1}{697}[48\sin(4x+3) + 63\cos(4x+3]]$

Exercise 4(a). Solve the following differential equations,

(1) $(D^2 - 1)y = 2x^4 - 3x + 1$ (2) $y'' + 2y' + 2y = x$ (3) $y'' + y' - 2y = x + \sin x$

(4) $\dfrac{d^3y}{dx^3} + \dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} = e^{2x} + x + x^2$ (5) $(D^2 + 4D + 4)y = x^2, y(0) = 0, y'(0) = 0$

Ans: (1) $y = (c_1 e^x + c_2 e^{-x}) - (2x^4 + 24x^2 - 3x + 49)$, (2) $y = c_1 e^{-x}\cos(x + c_2) + \dfrac{1}{2}(x-1)$

(3) $y = c_1 e^x + c_2 e^{-2x} - \dfrac{1}{10}(\cos x + 3\sin x) - \dfrac{1}{4}(2x+1)$

(4) $y = c_1 + \left\{c_2 \cos\left(\dfrac{\sqrt{3}}{2}x\right) + c_3 \sin\left(\dfrac{\sqrt{3}}{2}x\right)\right\}e^{-\frac{x}{2}} + \dfrac{1}{14}e^{2x} + \dfrac{x^3}{3} - \dfrac{x^2}{2} - x$

(5) $y = -\dfrac{3}{8}(1+2x)e^{-2x} + \dfrac{1}{8}(2x^2 + 3 - 4x)$

Exercise 4(b). Solve the following differential equations

(1) $\dfrac{d^2y}{dx^2} - 3\dfrac{dy}{dx} + 2y = xe^{3x} + \sin 2x$, (2) $\dfrac{d^2y}{dx^2} - 2\dfrac{dy}{dx} + 2y = x + e^x \cos x$, (3) $(D-2)^2 y = e^{2x}\sin x$, $y(0) = 1$, $y'(0) = -1$, (4) $(D^2 - 2D + 1)y = x^2 e^{3x}$, (5) $(D^3 - 2D + 4)y = e^x \sin x$, (6) $(D^2 - 6D + 13)y = 8\,e^{3x}\sin 2x$,

(7) $\dfrac{d^2y}{dx^2} - 4\dfrac{dy}{dx} + 4y = 8x^2 e^{2x}\sin 2x$, (8) $\dfrac{d^2y}{dx^2} - 2\dfrac{dy}{dx} + y = xe^x \sin x$, (9) $\dfrac{d^3y}{dx^3} - 7\dfrac{dy}{dx} - 6y = (x+1)e^{2x}$,

(10) $\dfrac{d^2y}{dx^2} - 6\dfrac{dy}{dx} + 13y = 8e^{3x}\sin 4x + 2^x$, (11) $\dfrac{d^3y}{dx^3} - 3\dfrac{d^2y}{dx^2} + 3\dfrac{dy}{dx} - y = (x+1)e^x$,

(12) $\dfrac{d^4y}{dx^4} - y = \cos x \cosh x$ (13) $(D^2 - 1)y = xe^x \sin x$

Ans: (1) $y = c_1 e^x + c_2 e^{2x} + \dfrac{e^{3x}}{2}\left(x - \dfrac{3}{2}\right) + \dfrac{1}{20}(3\cos 2x - \sin 2x)$

(2) $y = e^x(c_1 \cos x + c_2 \sin x) + (1/2)(x+1) + (1/2)xe^x \sin x$ (3) $y = (1-2x)e^{2x} - e^{2x}\sin x$

(4) $y = (c_1 + c_2 x)e^x + (2x^2 - 4x + 3)\dfrac{e^{3x}}{8}$ (5) $y = c_1 e^{-2x} + (c_2 \cos x + c_3 \sin x)e^x + (3\cos x + \sin x)\dfrac{xe^x}{20}$

(6) $y = e^{3x}(c_1 \cos 2x + c_2 \sin 2x - 2x\cos 2x)$ (7) $y = e^{2x}[c_1 + c_2 x + (3 - 2x^2)\sin 2x - 4x\cos 2x]$

(8) $y = (c_1 + c_2 x)e^x - e^x(x\sin x + 2\cos x)$ (9) $y = c_1 e^{-x} + c_2 e^{-2x} + c_3 e^{3x} - \dfrac{e^{2x}}{12}\left(x + \dfrac{17}{12}\right)$

(10) $y = e^{3x}(c_1\cos 2x + c_2\sin 2x) - (2/3)e^{3x}\sin 4x + \dfrac{2^x}{(\log 2)^2 - 6(\log 2) + 13}$

(11) $y = (c_1 + c_2 x + c_3 x^2)e^x + \dfrac{e^x x^4}{24} + \dfrac{e^x x^3}{6}$ (12) $y = c_1 e^x + c_2 e^{-x} + c_3\cos x + c_4\sin x - (1/5)\cos x\cosh x$

(13) $y = c_1 e^x + c_2 e^{-x} + \dfrac{1}{25}e^x[(14-5x)\sin x - 2(5x+1)\cos x]$

Exercise 5. Cauchy-Euler Homogeneous Equations: Solve the following differential equations,

(1) $x^2\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} + 4y = \cos(\log x) + x\sin(\log x)$ (2) $x^3\dfrac{d^3y}{dx^3} + 2x^2\dfrac{d^2y}{dx^2} + 2y = 10\left(x + \dfrac{1}{x}\right)$

(3) $x^2\dfrac{d^2y}{dx^2} - 2x\dfrac{dy}{dx} + 2y = \log^2 x - \log x^2$ (4) $x^2D^2 - 4xD + 4y = 4x^2 - 6x^3$, $y(2) = 4,\ y'(2) = -1$

(5) $x^2\dfrac{d^3y}{dx^3} + 3x\dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} = x^2\log x$ (6) $x^2\dfrac{d^2y}{dx^2} + 4x\dfrac{dy}{dx} + 2y = e^x$ (7) $x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = (\log x)(\sin\log x)$

(8) $x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} - y = x^3e^x$

Ans: (1) $y = x[c_1\cos(\sqrt{3}\log x) + c_2\sin(\sqrt{3}\log x)] + (1/13)[3\cos(\log x) - 2\sin(\log x)] + (1/2)x\sin\log x$

(2) $y = c_1x^{-1} + c_2x\cos(\log x + c_3) + 5x + 2x^{-1}\log x$

(3) $y = c_1x + c_2x^2 + \dfrac{1}{2}(\log^2 x + \log x) + \dfrac{1}{4}$ (4) $y = \dfrac{5}{3}x - 2x^2 + 3x^3 - \dfrac{23}{24}x^4$

(5) $y = [c_1 + c_2\log x + c_3(\log x)^2] - \dfrac{x^3}{27}(1 - \log x)$ (6) $y = \dfrac{c_1}{x} + \dfrac{c_2}{x^2} + \dfrac{e^x}{x^2}$

(7) $y = c_1\cos(\log x) + c_2\sin(\log x) + \dfrac{\log x}{4}[\sin(\log x) - (\log x)\cos(\log x)]$

(8) $y = c_1x + \dfrac{c_2}{x} + \left(x - 3 + \dfrac{3}{x}\right)e^x$

Exercise 6. Solve the following differential equations,

(1) $(3x+2)^2\dfrac{d^2y}{dx^2} + 3(3x+2)\dfrac{dy}{dx} - 36y = 3x^2 + 4x + 1$ (2) $(1+x)^2\dfrac{d^2y}{dx^2} + (1+x)\dfrac{dy}{dx} + y = 4\cos\log(1+x)$

(3) $(1+x)^2\dfrac{d^2y}{dx^2} + (1+x)\dfrac{dy}{dx} = (2x+3)(2x+4)$ (4) $(1+2x)^2\dfrac{d^2y}{dx^2} - 6(1+2x)\dfrac{dy}{dx} + 16y = 8(1+2x)^2$

(5) $(x+a)^2\dfrac{d^2y}{dx^2} - 4(x+a)\dfrac{dy}{dx} + 6y = x$

Ans: (1) $y = c_1(3x+2)^2 + c_2(3x+2)^{-2} + (1/108)[(3x+2)^2\log(3x+2) + 1]$

(2) $y = c_1\cos\{\log(1+x) + c_2\} + 2\log(1+x)\sin\log(1+x)$

(3) $y = c_1 + c_2\log(1+x) + (1+x)^2 + 6(x+1) + [\log(1+x)]^2$

(4) $y = (1+2x)^2[c_1 + c_2\log(1+2x) + \{\log(1+2x)\}^2]$

(5) $y = c_1(x+a)^2 + c_2(x+a)^3 + (1/2)(x+a) - (1/6)a$

Exercise 7. Solve the following Simultaneous differential equations,

1. $\dfrac{dx}{dt} = 3x + 8y$, $\dfrac{dy}{dt} = -x - 3y$ with $x(0) = 6,\ y(0) = -2$

   Ans: $x = 2e^{-t} + 4e^t$, $y = -e^{-t} - e^t$

2. $\dfrac{dx}{dt} + 4x + 3y = t$, $\dfrac{dy}{dt} + 2x + 5y = e^t$

Ans: $x=\frac{5}{14}t-\frac{3c_1}{2}e^{-2t}+c_2e^{-7t}-\frac{1}{8}e^t+\frac{5}{14}t-\frac{31}{196}$, $y=c_1e^{-2t}+c_2e^{-7t}+\frac{5}{24}t-\frac{1}{7}\left(t-\frac{9}{14}\right)$

3. $\frac{dx}{dt}+5x-2y=t$, $\frac{dy}{dt}+2x+y=0$, given that $x=y=0$ when $t=0$

   Ans: $x=-\frac{1}{27}(1+6t)e^{-3t}+\frac{1}{27}(1+3t)$, $y=-\frac{2}{27}(2+3t)e^{-3t}+\frac{2}{27}(2-3t)$

4. $\frac{dx}{dt}+2\frac{dy}{dt}-2x+2y=3e^t$, $3\frac{dx}{dt}+\frac{dy}{dt}+2x+y=4e^{2t}$

   Ans: $x=(1/2)e^{2t}-(3/11)e^t+c_1e^{-(6/5)t}$, $y=(15/22)e^t-8c_1e^{-(6/5)t}+c_2e^{-t}$

5. The equations of motion of a particle are given by $\frac{dx}{dt}+\omega y=0$, $\frac{dy}{dt}-\omega x=0$. Find the path of the particle and show that it is a circle.

   Ans: The path is given by, $x=c_1\cos\omega t+c_2\sin\omega t$, $y=c_1\sin\omega t-c_2\cos\omega t$

Exercise 8. Solve the following Simultaneous differential equations,

1. $x'+2y+\sin t=0$, $y'-2x-\cos t=0$, given that $x=0$ and $y=1$ when $t=0$.

   Ans: $x=\cos 2t-\sin 2t-\cos t$, $y=\cos 2t+\sin 2t-\sin t$

2. $x'+y=\sin t$, $y'+x=\cos t$, given that $x=2$ and $y=0$ when $t=0$.

   Ans: $x=e^{-t}+e^t$, $y=e^{-t}-e^t+\sin t$

3. $\frac{d^2x}{dt^2}+\frac{dy}{dt}+3x=e^{-t}$, $\frac{d^2y}{dt^2}-4\frac{dx}{dt}+3y=\sin 2t$ **Ans:** $x=c_1\cos t+c_2\sin t+c_3\cos 3t+c_4\sin 3t+\frac{e^{-t}}{5}+\frac{2}{15}\cos 2t$

   $y=\frac{1}{15}\sin 2t+c_5\sin t+c_6\cos t+c_7\sin 3t+c_8\cos 3t-\frac{1}{5}e^{-t}$

4. $\frac{dx}{dt}+\frac{dy}{dt}-2y=2\cos t-7\sin t$, $\frac{dx}{dt}-\frac{dy}{dt}+2x=4\cos t-3\sin t$

   Ans: $x=c_1e^{\sqrt{2}t}+c_2e^{-\sqrt{2}t}+3\cos t$, $y=\left(1+\frac{1}{\sqrt{2}}\right)c_1e^{\sqrt{2}t}+\left(1-\frac{1}{\sqrt{2}}\right)c_2e^{-\sqrt{2}t}+2\sin t+c_3$

5. $\frac{d^2x}{dt^2}+y=\sin t$, $\frac{d^2y}{dt^2}+x=\cos t$

   Ans: $x=c_1e^t+c_2e^{-t}+c_3\cos t+c_4\sin t+(t/4)(\sin t-\cos t)$

   $y=-c_1e^t-c_2e^{-t}+c_3\cos t+c_4\sin t+(1/4)(t+2)(\sin t-\cos t)$

Exercise 9. Solve the following differential equations,

(1) $x\frac{d^2y}{dx^2}-(2x-1)\frac{dy}{dx}+(x-1)y=0$ (2) $x\frac{d^2y}{dx^2}+(1-x)\frac{dy}{dx}-y=e^x$ (3) $x^2\frac{d^2y}{dx^2}-(x^2+2x)\frac{dy}{dx}+(x+2)y=x^3e^x$

(4) Solve $\frac{d^2y}{dx^2}-4x\frac{dy}{dx}+(4x^2-2)y=0$ given that $y=e^{x^2}$ is an integral included in the complimentary function. (5) Solve $\sin^2 x\frac{d^2y}{dx^2}=2y$ given that $y=\cot x$ is an integral included in the complimentary function. (6) Solve $(x\sin x+\cos x)\frac{d^2y}{dx^2}-x\cos x\frac{dy}{dx}+y\cos x=0$ given that $y=x$ is an integral included in the complimentary function.

Ans: (1) $y=e^x(c_1\log x+c_2)$ (2) $y=e^x\log x+c_1e^x\int\frac{e^{-x}}{x}dx+c_2e^x$ (3) $y=c_1xe^x+c_2x+x(x-1)e^x$

(4) $y=(c_1+c_2x)e^{x^2}$ (5) $y=c_1(1-x\cot x)+c_2\cot x$ (6) $y=-c_1\cos x+c_2x$

Exercise 10 Solve the following differential equations by Normal form (Removal of first derivative)

(1) $\frac{d^2y}{dx^2} - 4x\frac{dy}{dx} + (4x^2 - 3)y = e^{x^2}$ (2) $\frac{d^2y}{dx^2} - \frac{2}{x}\frac{dy}{dx} + \left(1 + \frac{2}{x^2}\right)y = xe^x$ (3) $y'' - 4xy' + (4x^2 - 1)y = -3e^{x^2}\sin 2x$

(4) $\left(\frac{d^2y}{dx^2} + y\right)\cot x + 2\left(\frac{dy}{dx} + y\tan x\right) = \sec x$ (5) $y'' - \frac{2}{x}y' + \left(1 + \frac{2}{x^2}\right)y = xe^x$ (6) $\frac{d}{dx}\left(\cos^2 x\frac{dy}{dx}\right) + (\cos^2 x)y = 0$

Ans:(1) $y = e^{x^2}(c_1e^x + c_2e^{-x} - 1)$ (2) $y = x\{c_1\cos x + c_2\sin x + (1/2)e^x\}$

(3) $y = e^{x^2}\{c_1\cos x + c_2\sin x + \sin 2x\}$ (4) $y = (c_1x + c_2)\cos x + \frac{1}{2}(\sin x - x\cos x)$

(5) $y = x\left(c_1\cos x + c_2\sin x + \frac{e^x}{2}\right)$ (6) $y = (c_1\cos\sqrt{2}\,x + c_2\sin\sqrt{2}\,x)\sec x$

Exercise 11 Solve the following differential equations by changing independent variable

(1) $\frac{d^2y}{dx^2} - \frac{1}{x}\frac{dy}{dx} + 4x^2y = x^4$ (2) $x\frac{d^2y}{dx^2} + (4x^2 - 1)\frac{dy}{dx} + 4x^3y = 2x^3$ (3) $\frac{d^2y}{dx^2} + \cot x\frac{dy}{dx} + 4y\,\mathrm{cosec}^2 x = 0$

(4) $\cos x\frac{d^2y}{dx^2} + \sin x\frac{dy}{dx} - 2y\cos^3 x = 2\cos^5 x$ (5) $\frac{d^2y}{dx^2} + (\tan x - 1)^2\frac{dy}{dx} - n(n-1)y\sec^4 x = 0$

(6) $x^4\frac{d^2y}{dx^2} + 2x^3\frac{dy}{dx} + n^2y = 0$ (7) $(1+x)^2\frac{d^2y}{dx^2} + (1+x)\frac{dy}{dx} + y = 4\cos\log(1+x)$

(8) $\frac{d^2y}{dx^2} + (3\sin x - \cot x)\frac{dy}{dx} + (2\sin^2 x)y = e^{-\cos x}\sin^2 x$

Ans: (1) $y = c_1\cos x^2 + c_2\sin x^2 + \frac{x^2}{4}$ (2) $y = (c_1 + c_2x^2)e^{-x^2} + (1/2)$

(3) $y = c_1\cos\{2\log\tan(x/2)\} + c_2\sin\{2\log\tan(x/2)\}$ (4) $y = c_1e^{\sqrt{2}\sin x} + c_2e^{-\sqrt{2}\sin x} + \sin^2 x$

(5) $y = c_1e^{-n\tan x} + c_2e^{(n-1)\tan x}$ (6) $y = c_1\cos\left(-\frac{n}{x}\right) + c_2\sin\left(-\frac{n}{x}\right)$

(7) $y = c_1\cos\{\log(1+x) + c_2\} + 2\log(1+x)\sin\log(1+x)$ (8) $y = c_1e^{\cos x} + c_2e^{2\cos x} + \frac{e^{-\cos x}}{6}$

Exercise 12 Solve the following differential equations by the method of variation of parameter

(1) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = \frac{e^x}{1+e^x}$ (2) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = xe^x\sin x$ (3) $(D^2 - 3D + 2)y = xe^x + 2x$

(4) $(D^2 - 1)y = 2(1 - e^{-2x})^{-\frac{1}{2}}$ (5) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = e^x\sin x$

Ans: (1) $y = c_1e^x + c_2e^{2x} + e^x\log(e^{-x} + 1) - e^x + e^{2x}\log(e^{-x} + 1)$ (2) $y = (c_1 + c_2x)e^x - e^x(x\sin x + 2\cos x)$

(3) $y = c_1e^x + c_2e^{2x} - (x^2/2)e^x - xe^{-x} + x + 3/2$ (4) $y = c_1e^x + c_2e^{-x} - e^x\sin^{-1}(e^{-x}) - (e^{2x} - 1)^{\frac{1}{2}}e^{-x}$

(5) $y = c_1 + c_2e^{2x} - \frac{e^x}{2}\sin x$

Exercise 13 Solve the following differential equations by the method of variation of parameter-

(1) $(D^2 + a^2)y = \sec ax$, (2) $(D^2 + a^2)y = \cos ax$, (3) $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = x^2e^x$,

(4) $y'' + y = \tan x$, (5) $y'' + 3y' + 2y = e^{e^x}$, (6) $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = x^3e^x$,

(7) If $y_1 = x$ and $y_2 = \frac{1}{x}$ are two linearly independent solutions of the differential equation $x^2y'' + xy' - y = x,\ x \neq 0$, find the general solution of the equation.

Ans: (1) $y = c_1 \cos ax + c_2 \sin ax + \frac{x \sin ax}{a} + \frac{\cos ax \log \cos ax}{a^2}$

(2) $y = c_1 \cos ax + c_2 \sin ax + \frac{\cos ax \cos 2ax}{4a^2} + \frac{\sin ax}{2a}\left(x + \frac{\sin 2ax}{2a}\right)$ (3) $y = c_1 x + \frac{c_2}{x} + e^x - \frac{e^x}{x}$

(4) $y = c_1 \cos x + c_2 \sin x - \cos x.\log(\sec x + \tan x)$, (5) $y = c_1 e^{-x} + c_2 e^{-2x} - e^{e^x} e^{-2x}$

(6) $y = c_1 \frac{1}{x} + c_2 x + e^x\left(x + \frac{3}{x} - 3\right)$, (7) $y = \left(c_1 - \frac{1}{4}\right)x + \frac{1}{x}c_2 + \frac{x}{2}\log x$

Exercise 14.

1. A particle of unit mass falls under gravity in a resisting medium whose resistance varies with velocity. Find the relation between distance and velocity if initially the particle starts from rest.

   Ans: $s = \frac{g}{k^2}\log\left(\frac{g}{g - kv}\right) - \frac{v}{k}$

2. An R-L circuit has an emf $4\sin t$ volts, a resistence of 100 ohms, an inductance of 4 henries with no initial current. Find the current at any time '*t*'. **Ans:** $i = \frac{1}{626}(e^{-25t} - \cos t + 25\sin t)$

3. An alternative emf $E \sin \omega t$ is applied to an inductance L and capacitance C in series. Show that the current in the circuit is $\frac{E\omega}{(n^2 - \omega^2)L}(\cos \omega t - \cos nt)$, where $n^2 = \frac{1}{LC}$.

4. The differential equation $\frac{d^2x}{dt^2} + 2k\frac{dx}{dt} + n^2x = 0, (k < n)$ represents the damped harmonic oscillations of a particle. Solve this equation and show that the ratio of the amplitude of any oscillation to that of the preceding one is constant i.e. its amplitude form a geometrical progression. **Ans:** ratio = e^{-k} (a constant)

5. The deflection of a strut of length '*l*' with one end (x=0) built in and the other supported and subjected to end thrust *P*, satisfies the equation $\frac{d^2y}{dx^2} + a^2y = \frac{a^2R}{P}(l - x)$. Prove that deflection curve is $y = \frac{R}{P}\left(\frac{\sin ax}{a} - l\cos ax + l - x\right)$, where $al = \tan al$.